

#1. (4.2) $p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$

(4.3) $\frac{p(X)}{1-p(X)} = e^{\beta_0 + \beta_1 X}$

$$\frac{p(X)}{1-p(X)} = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \cdot \left(1 - \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}\right) =$$

$$\frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \cdot \left(\frac{1 + e^{\beta_0 + \beta_1 X} - e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}\right) =$$

$$\frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \cdot (1 + e^{\beta_0 + \beta_1 X} - e^{\beta_0 + \beta_1 X}) = e^{\beta_0 + \beta_1 X} \quad \square$$

#2. (4.17) $p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(x-\mu_k)^2)}{\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(x-\mu_l)^2)}$

(4.18) $\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$

Prove that $\arg\max(p_k(x)) = \arg\max(\delta_k(x))$
 $\neq \arg\max = \max$

Because the denominator is the same for all $p_k(x)$, it won't change the $\arg\max(f(x))$, so it can be dropped. Furthermore, $\log$ transformation only scales $f(x)$, where relative ranking remains intact, thus, still,
 $\arg\max(f(x)) = \arg\max(\log(f(x)))$

$$\arg\max(p_k(x)) = \log\left(\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(x-\mu_k)^2)\right) =$$

$$\arg\max\left(\log\left(\frac{\pi_k}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2} \frac{(x-\mu_k)^2}{\sigma^2}\right) =$$

terms that don't depend on k are constants
 and thus don't change the $\arg\max(f(x))$

$$\arg\max\left(\log(\pi_k) - \frac{1}{2\sigma^2}(x^2 - 2x\mu_k + \mu_k^2)\right) =$$

$$\arg\max\left(\log(\pi_k) + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}\right) =$$

$$\arg\max\left(x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)\right) = \arg\max(\delta_k(x)) \quad \square$$

#3. (4.15) $P(Y=k | X=x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$

(4.16) $f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$

$$\arg\max(P(Y=k | X=x)) = \arg\max(\log(\pi_k f_k(x))) =$$

$$\arg\max\left(\log(\pi_k) + \log\left(\frac{1}{\sqrt{2\pi}\sigma_k}\right) - \log(\sigma_k)\right) =$$

$$- \frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} =$$

$$\arg\max\left(\log(\pi_k) - \log(\sigma_k) - \frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2}\right) =$$

thus, $\delta_k(x) = \log(\pi_k) - \log(\sigma_k) - \frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2}$
 for QDA w $p=1$ and K classes

the $\delta_k(x)$ involves x^2 , making the Bayes's classifier / boundary non-linear, specifically quadratic. $\square$

#4. a) About 1/10 of all observations will be used to make a prediction given $X \sim \text{Uniform}$

b) $X_1, X_2 \sim \text{Uniform}$
 X_1, X_2 independent
 $P(Y|X_1, X_2)$
 $P(X_1) = P(X_2) = 0.1$
 $P(X_1 \cap X_2) = P(X_1) \cdot P(X_2) = 0.01$

Geometrically, KNN looks for neighbours that fall under both $[0.55, 0.65] \cap [0.3, 0.4]$ intersects and not union of $X_1 \cup X_2$.

c) $P(X_1, n \dots \cap X_{100}) = 0.1^{100}$

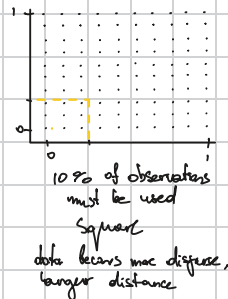
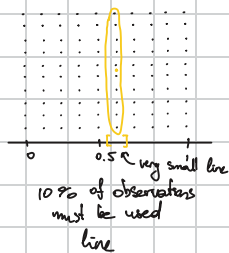
d) The more predictors we have, the more difficult it is for a KNN model to find neighbours in all dimensions to make a correct prediction. This means that only a very few observations will be used, making it hard to capture any trends that the dataset might contain that otherwise could have been used usefully.
(fixed distance $\uparrow$ for parts a, b, c, d)

e) $p = 1, 2, 100$ (fixed $k \downarrow$ for part e)

When $p=1$, 10% of the nearest values are used. Given the space is a line of length 1 with uniform distribution, the hypercube length therefore is 0.1.

When $p=2$, the space becomes a square. Area must contain 10% of the observations, thus $A = 0.1 = l^2 = \sqrt{0.1}$, the hypercube length is 0.32.

When $p=3$, the space becomes a cube in 100D whose volume contains 10% of observations. Thus $V = 0.1 = l^{100} = \sqrt[100]{0.1} \approx 0.377$



#5. a) The QDA will perform better on the training set due to overfitting the linear data. However, the LDA will perform better on the test set because it will approximate the Bayes linear boundary better.

b) In case of non-linear data, the QDA will outperform LDA in both training and testing sets, following the Bayes boundary closer.

c) As n increases, the variance of the samples will influence QDA less and less and QDA's accuracy might increase. However, LDA will still outperform QDA on non-linear sets.

d) False. QDA will perform worse than LDA on testing sets in linear data because, as it gets trained, the regression has more room to overfit the training set and perform worse on the test set.

#16. $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.05, \hat{\beta}_2 = 1$

X_1 = Hours studied
 X_2 = GPA
 Y = receive an A

$$p(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2}}$$

a) $p(X) = \frac{\exp(-6 + 0.05 \cdot 40 + 3.5)}{1 + \exp(-6 + 0.05 \cdot 40 + 3.5)} \approx 0.38 \approx 38\%$

b) $0.5 = \frac{\exp(-6 + 0.05 \cdot X_1 + 3.5)}{1 + \exp(-6 + 0.05 \cdot X_1 + 3.5)}$

$$0.5 + 0.5(\exp(0.05 X_1 - 2.5)) = \exp(0.05 X_1 - 2.5)$$

$$0.5 = 0.5 \exp(0.05 X_1 - 2.5)$$

$$\exp(0.05X_1 - 2.5) = 1$$

$$0.05X_1 - 2.5 = \ln(1)$$

$$0.05X_1 = 2.5$$

$$X_1 = 50 \text{ hours}$$

#7. LDA, $p=1$, $\hat{\sigma}^2=6$, $\mu_k=10$, $\mu_0=0$
 $x=4$, $\pi_k=0.8$, $\pi_0=0.2$

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu_k)^2\right)}{\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu_l)^2\right)} \quad (4.17)$$

$$p_k(x=4) = \frac{0.8 \frac{1}{6 \cdot \sqrt{2\pi}} \exp\left(-\frac{(4-10)^2}{2 \cdot 36}\right)}{0.8 \frac{1}{6 \cdot \sqrt{2\pi}} \exp\left(-\frac{(4-10)^2}{2 \cdot 36}\right) + 0.2 \frac{1}{6 \cdot \sqrt{2\pi}} \exp\left(-\frac{(4-0)^2}{2 \cdot 36}\right)}$$

$$\rightarrow p_k(x=4) = 0.75 \approx 75\%$$

#8. Train-test split: 50% / 50%

Logistic Regression

Training Error 20%

Testing Error 30%

KNN (k=1)

Avg. error 18%

KNN's training error must be around 0 due to very low h (of 1). Thus, $\frac{2+1}{2}=1.5 \Rightarrow$ testing error should be close to 36%. Such model struggles w/ overfitting. Because the logistic regression has lower testing error, it should be chosen instead.

#9. Odds,

$$a) \text{ odds} = 0.37 = \frac{p(X)}{1-p(X)}$$

$$0.37 - 0.37 p(X) = p(X)$$

$$1.37 p(X) = 0.37$$

$$p(X) = \frac{0.37}{1.37} \approx 0.27$$

27% of the people will default

$$b) p(X) = 16\%$$

$$\text{odds} = \frac{p(X)}{1-p(X)} = \frac{0.16}{1-0.16} \approx 0.19$$

the odds of default are 0.19

#10. By setting $\downarrow$ to 0, you'll find the prediction boundary

$$\log\left(\frac{P(Y=k|X=x)}{P(Y=K|X=x)}\right) = \log\left(\frac{\pi_k f_k(x)}{\pi_K f_K(x)}\right)$$

$$\log\left(\frac{\frac{\pi_k}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu_k)^2\right)}{\frac{\pi_K}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu_K)^2\right)}\right)$$

$$\log(\pi_k \exp(-\frac{1}{2\sigma^2}(x-\mu_k)^2)) - \log(\pi_K \exp(-\frac{1}{2\sigma^2}(x-\mu_K)^2))$$

$$\log(\pi_k) - \frac{1}{2\sigma^2}(x-\mu_k)^2 - \log(\pi_K) + \frac{1}{2\sigma^2}(x-\mu_K)^2$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) + \frac{1}{2\sigma^2}((x-\mu_K)^2 - (x-\mu_k)^2)$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) + \frac{1}{2\sigma^2} \left(x^2 - 2\mu_k x + \mu_k^2 - (x^2 - 2\mu_K x + \mu_K^2) \right)$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) + \frac{1}{2\sigma^2} \left(\cancel{x^2} - 2\mu_k x + \mu_k^2 - \cancel{x^2} + 2\mu_K x - \mu_K^2 \right)$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) - \frac{x\mu_k}{\sigma^2} + \frac{\mu_k^2}{2\sigma^2} + \frac{x\mu_K}{\sigma^2} - \frac{\mu_K^2}{2\sigma^2}$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) + \frac{x\mu_K}{\sigma^2} - \frac{x\mu_k}{\sigma^2} + \frac{\mu_K^2}{2\sigma^2} - \frac{\mu_k^2}{2\sigma^2}$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) + \frac{x}{\sigma^2} (\mu_K - \mu_k) + \frac{1}{2\sigma^2} (\mu_K^2 - \mu_k^2)$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) - \frac{(\mu_k - \mu_K)x}{\sigma^2} + \frac{1}{2\sigma^2} (\mu_K^2 - \mu_k^2)$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) + \frac{1}{2\sigma^2} (\mu_K^2 - \mu_k^2) - \frac{(\mu_k - \mu_K)x}{\sigma^2}$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) + \frac{\mu_K - \mu_k}{\sigma^2} \left(\frac{\mu_K + \mu_k}{2} - x \right)$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) - \frac{\mu_k - \mu_K}{\sigma^2} \left(\frac{\mu_k + \mu_K}{2} - x \right)$$

$$\log\left(\frac{\pi_k}{\pi_K}\right) - \frac{\mu_k^2 - \mu_K^2}{2\sigma^2} + \frac{\mu_k - \mu_K}{\sigma^2} x$$

when $p=1$, $a_k + \sum_{j=1}^p b_{kj} x_j$ becomes $a_k + b_{k1} x_1$

where,

$$a_k = \log\left(\frac{\pi_k}{\pi_K}\right) - \frac{\mu_k^2 - \mu_K^2}{2\sigma^2} \quad b_{k1} = \frac{\mu_k - \mu_K}{\sigma^2}$$

#11. Work out the detailed forms of a_k , b_{kj} , and b_{kjl} in (4.33). Your answer should involve π_k , π_K , μ_k , μ_K , Σ_k , and Σ_K .

$$f(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right). \quad (4.23)$$

$$\log\left(\frac{\Pr(Y=k|X=x)}{\Pr(Y=K|X=x)}\right) =$$

$$= \log\left(\frac{\frac{\pi_k}{(2\pi)^{p/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k)\right)}{\frac{\pi_K}{(2\pi)^{p/2} |\Sigma_K|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_K)^T \Sigma_K^{-1}(x-\mu_K)\right)}\right)$$

$$= \log\left(\frac{\pi_k}{(2\pi)^{p/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k)\right)\right) - \log\left(\frac{\pi_K}{(2\pi)^{p/2} |\Sigma_K|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_K)^T \Sigma_K^{-1}(x-\mu_K)\right)\right)$$

$$= \log\left(\frac{\pi_k}{(2\pi)^{p/2} |\Sigma_k|^{1/2}}\right) - \frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k) - \log\left(\frac{\pi_K}{(2\pi)^{p/2} |\Sigma_K|^{1/2}}\right) + \frac{1}{2}(x-\mu_K)^T \Sigma_K^{-1}(x-\mu_K)$$

$$= \log(\pi_k) - \log((2\pi)^{p/2} |\Sigma_k|^{1/2}) - \log(\pi_K) + \log((2\pi)^{p/2} |\Sigma_K|^{1/2}) - \frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k) + \frac{1}{2}(x-\mu_K)^T \Sigma_K^{-1}(x-\mu_K) = \frac{1}{2} + \frac{1}{2} \downarrow$$

$$= \log\left(\frac{\pi_k}{\pi_K} \frac{|\Sigma_K|^{1/2}}{|\Sigma_k|^{1/2}}\right) + \frac{1}{2} \left(\mu_K^T \Sigma_K^{-1} \mu_K - \mu_k^T \Sigma_k^{-1} \mu_k \right) + x^T \left(\frac{1}{2} (\Sigma_K^{-1} - \Sigma_k^{-1}) \right) x$$

$$\log\left(\frac{\Pr(Y=k|X=x)}{\Pr(Y=K|X=x)}\right) = a_k + \sum_{j=1}^p b_{kj} x_j + \sum_{j=1}^p \sum_{l=1}^p c_{kjl} x_j x_l, \quad (4.33)$$

$$l = \log(\bar{y}_k) - \log((2.5)^{p/2} |\Sigma_k|^{1/2}) - \log(\bar{y}_k) + \log((2.5)^{p/2} |\Sigma_k|^{1/2})$$

$$= \log\left(\frac{\bar{y}_k}{\bar{y}_k}\right) - \log((2.5)^{p/2}) - \log(|\Sigma_k|^{1/2}) + \log((2.5)^{p/2}) + \log(|\Sigma_k|^{1/2})$$

$$= \log\left(\frac{\bar{y}_k}{\bar{y}_k}\right) + \log\left(\frac{|\Sigma_k|^{1/2}}{|\Sigma_k|^{1/2}}\right) = \log\left(\frac{\bar{y}_k}{\bar{y}_k} \frac{|\Sigma_k|^{1/2}}{|\Sigma_k|^{1/2}}\right)$$

$$= \frac{1}{2} \left((x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) - (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right)$$

$$= \frac{1}{2} \left(x^T \Sigma_k^{-1} x - 2x^T \Sigma_k^{-1} \mu_k + \mu_k^T \Sigma_k^{-1} \mu_k - x^T \Sigma_k^{-1} x + 2x^T \Sigma_k^{-1} \mu_k - \mu_k^T \Sigma_k^{-1} \mu_k \right)$$

$$= \frac{1}{2} \left(x^T \Sigma_k^{-1} x - x^T \Sigma_k^{-1} x - 2x^T \Sigma_k^{-1} \mu_k + 2x^T \Sigma_k^{-1} \mu_k + \mu_k^T \Sigma_k^{-1} \mu_k - \mu_k^T \Sigma_k^{-1} \mu_k \right)$$

$$= \underbrace{x^T \left(\frac{1}{2} (\Sigma_k^{-1} - \Sigma_k^{-1}) \right) x}_{x^2} + \underbrace{x^T (\Sigma_k^{-1} \mu_k - \Sigma_k^{-1} \mu_k)}_x + \underbrace{\frac{1}{2} (\mu_k^T \Sigma_k^{-1} \mu_k - \mu_k^T \Sigma_k^{-1} \mu_k)}_{\text{constant}}$$

derivation:

$$(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) = x^T \Sigma_k^{-1} x - x^T \Sigma_k^{-1} \mu_k - \mu_k^T \Sigma_k^{-1} x + \mu_k^T \Sigma_k^{-1} \mu_k$$

$$\Sigma_k^{-1} = (\Sigma_k^{-1})^T \rightarrow x^T \Sigma_k^{-1} \mu_k = \mu_k^T \Sigma_k^{-1} x = \text{same scalar}$$

$$(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) = x^T \Sigma_k^{-1} x - 2x^T \Sigma_k^{-1} \mu_k + \mu_k^T \Sigma_k^{-1} \mu_k$$

$$\begin{matrix} x & \Sigma^{-1} & \mu \\ \begin{bmatrix} a \\ b \end{bmatrix} & \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix} & \begin{bmatrix} d \\ e \end{bmatrix} \\ 2 \times 1 & 2 \times 2 & 2 \times 1 \end{matrix}$$

$$x^T \Sigma_k^{-1} \mu_k \approx \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix} \begin{bmatrix} d \\ e \end{bmatrix} = \begin{bmatrix} a+bc & ac+b \end{bmatrix} \begin{bmatrix} d \\ e \end{bmatrix} = \begin{bmatrix} d(a+bc) & e(ac+b) \end{bmatrix} = \begin{bmatrix} ad+bcd & ace+be \end{bmatrix} \leftarrow \text{scalar}$$

$$\mu_k^T \Sigma_k^{-1} x \approx \begin{bmatrix} d & e \end{bmatrix} \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} d(ac+ec) & e(bc+e) \end{bmatrix} = \begin{bmatrix} ad+ace & bcd+be \end{bmatrix} \leftarrow \text{same scalars}$$

$$a_k = \log\left(\frac{\bar{y}_k}{\bar{y}_k} \frac{|\Sigma_k|^{1/2}}{|\Sigma_k|^{1/2}}\right) + \frac{1}{2} \left(\mu_k^T \Sigma_k^{-1} \mu_k - \mu_k^T \Sigma_k^{-1} \mu_k \right)$$

$$b_{kj} = (\Sigma_k^{-1} \mu_k - \Sigma_k^{-1} \mu_k)_j$$

$$b_{kj} = \left(\frac{1}{2} (\Sigma_k^{-1} - \Sigma_k^{-1}) \right)_{jl}$$

#12. a) log-odds of logistic regression: $\hat{\beta}_0 + \hat{\beta}_1 x$

b) log-odds of softmax logistic regression:

$$\underbrace{(\hat{\alpha}_{orange} - \hat{\alpha}_{apple})}_{||} + \underbrace{(\hat{\alpha}_{orange} - \hat{\alpha}_{apple})}_{||} x$$

$$c) \quad 2, \hat{\beta}_0 \quad \leftarrow \quad -1, \hat{\beta}_1$$

$$\text{if } \hat{\beta}_0 = 2, \hat{\beta}_1 = -1$$

$$d) \hat{\beta}_0 = (1.2 - 3) = -1.8$$

$$\hat{\beta}_1 = (-2 - 0.6) = -2.6$$

e) Because parameters for two models are the same, their results should match perfectly as well.